

Interdependence and the Gains from Trade

Martin J. Conyon

Bentley University

Required readings

The core text is N. Gregory Mankiw, *Principles of Microeconomics*.

- **Chapter 3** — Interdependence and the Gains from Trade.

Read the chapter alongside these slides.

1. **Interdependence** — why you consume goods made by strangers.
2. **The production possibilities frontier** — the workhorse diagram.
3. **Increasing opportunity cost** — why frontiers usually bend.
4. **Efficiency, unemployment, growth** — reading points on, inside and beyond the frontier.
5. **Absolute and comparative advantage.**
6. **Specialization and the gains.**
7. **The terms of trade.**
8. **Limits.**

PART 1

Interdependence

You depend on strangers

Every day you consume goods produced by people you will never meet.

- Your coffee was grown in Colombia.
- Your phone was assembled abroad from parts made in several countries.
- Your shirt was sewn from cotton grown on another continent.

None of these people know you. **None act out of kindness.**

Why does any of it reach you?

Because both sides gain.

That sounds obvious. It is not.

The surprising claim. Two countries can both gain from trade *even when one of them is better at producing everything*.

If one country is more productive at every good, what could the other possibly offer?

Where the idea comes from

David Ricardo — London stockbroker and Member of Parliament — in *On the Principles of Political Economy and Taxation*, 1817.

Adam Smith had already shown you should buy from abroad when the foreigner is cheaper. Ricardo went further.

His example: **England and Portugal, cloth and wine**. We keep his countries, his goods, and his famous numbers.

Active Learning 1

Before we start the analysis — what do you predict?

Portugal is better than England at making *both* cloth and wine.

- A. Portugal should produce both and England should produce nothing.
- B. Portugal gains from trade, England loses.
- C. Both gain from trade.
- D. Neither gains — there is nothing to trade.

60 seconds. Commit to an answer and remember it.

PART 2

The Production Possibilities Frontier

England and Portugal each produce cloth and wine using **labor alone**.

Hours of labor needed per unit		
	Cloth	Wine
England	100	120
Portugal	90	80

Read this carefully: these are costs. A bigger number means worse.

A warning about conventions

Ricardo writes **hours per unit**. Mankiw writes **units per hour**.

These are *reciprocals* of each other:

$$100 \text{ hours per cloth} \iff \frac{1}{100} = 0.01 \text{ cloth per hour}$$

Neither is wrong. But check which one you are looking at before you compare numbers.

How much labor does each country have?

Ricardo gave the productivity figures but never said how much labor each country has. We must supply that to draw anything.

England 3,600 hours. Portugal 2,880 hours. Think of ten workers against eight, each working the same time.

Chosen so every number below comes out whole. **We change only the workforce sizes, never Ricardo's productivity numbers** — those carry the economics.

Step 1 — write the frontier as an equation

Production possibilities frontier (PPF). The *maximum* combinations of the two goods a country can produce when it uses all of its resources *efficiently*, given its technology.

Labor on cloth plus labor on wine equals total labor. For England:

$$100 C + 120 W = 3600$$

Solve for C to put cloth on the vertical axis:

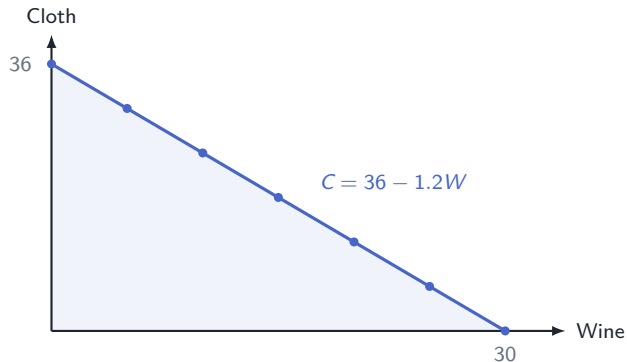
$$C = 36 - 1.2 W$$

Step 2 — plug the numbers in

England: $C = 36 - 1.2 W$		
Wine W	Substitute	Cloth C
0	$36 - 1.2(0)$	36
5	$36 - 1.2(5)$	30
10	$36 - 1.2(10)$	24
15	$36 - 1.2(15)$	18
20	$36 - 1.2(20)$	12
25	$36 - 1.2(25)$	6
30	$36 - 1.2(30)$	0

The ends: **36 cloth** if England makes no wine; **30 wine** if it makes no cloth.

Step 3 — draw the graph



Straight, because labor productivity is constant. The shaded area is everything England *can* produce.

What the slope means

The slope is -1.2 . Move along the frontier and each extra wine costs England **1.2 cloth**.

The slope of the frontier is the opportunity cost.

And it came straight from the labor data:

$$\frac{120 \text{ hours per wine}}{100 \text{ hours per cloth}} = 1.2$$

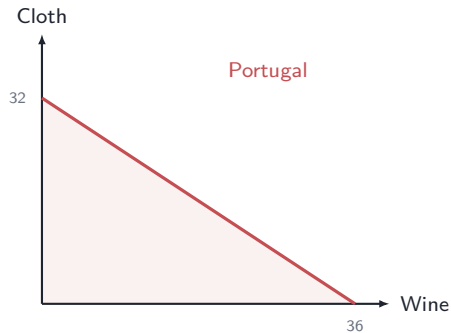
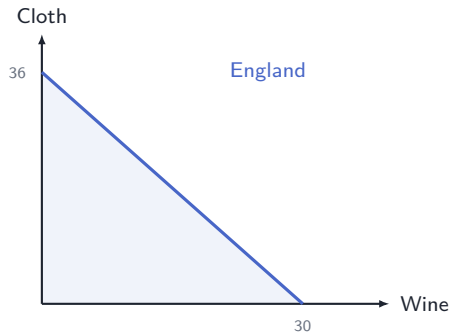
Now build Portugal's frontier.

Portugal: 90 hours per cloth, 80 hours per wine, **2,880 hours** of labor.

1. Write the labor constraint, then solve for C .
2. What is the maximum cloth? The maximum wine?
3. What is the slope, and what does it mean in words?

On paper. Four minutes.

The two frontiers, side by side



Portugal's line is flatter. Its wine is cheaper, in cloth.

PART 3

Increasing Opportunity Cost

Straight lines are the special case

Both frontiers so far are straight, because we assumed labor productivity is **constant** — every hour moved buys the same amount as the last.

Real economies are not like that.

Resources are not identical. The land best suited to wine is not the land best suited to cloth.

A second economy, with a bent frontier

An economy produces **food** and **clothing**. Its frontier is

$$C = 50 - 0.02 F^2$$

Same idea as before: it is an equation, so we tabulate it and plot it.

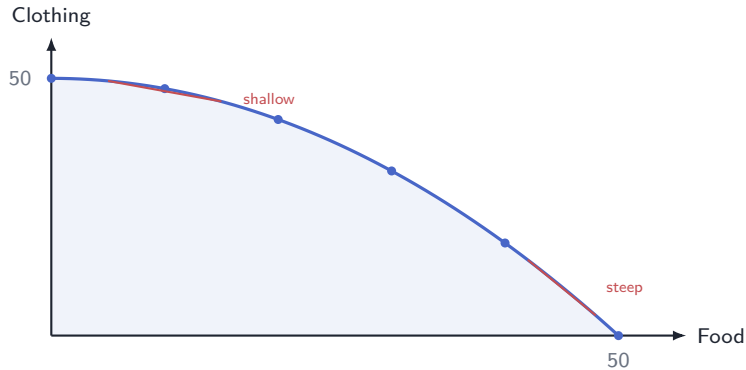
Watch what happens to the cost of each extra 10 units of food.

Plug the numbers in

Food F	$C = 50 - 0.02F^2$	Clothing C	Cost of the last 10 food
0	$50 - 0.02(0)$	50	—
10	$50 - 0.02(100)$	48	2
20	$50 - 0.02(400)$	42	6
30	$50 - 0.02(900)$	32	10
40	$50 - 0.02(1600)$	18	14
50	$50 - 0.02(2500)$	0	18

2, 6, 10, 14, 18. Each extra 10 food costs more clothing than the last.

The bowed frontier



The frontier bows outward. Its slope steepens as you move right.

Why frontiers bow

Start by producing only clothing. Move resources to food and you move the resources *best suited to food* first — cheap.

Keep going and you must move resources that were *good at clothing* — expensive.

Increasing opportunity cost is what specialized resources look like on a diagram.

Constant cost (a straight line) is the special case where all resources are interchangeable.

Active Learning 3

An economy's frontier is $C = 80 - 0.05 F^2$.

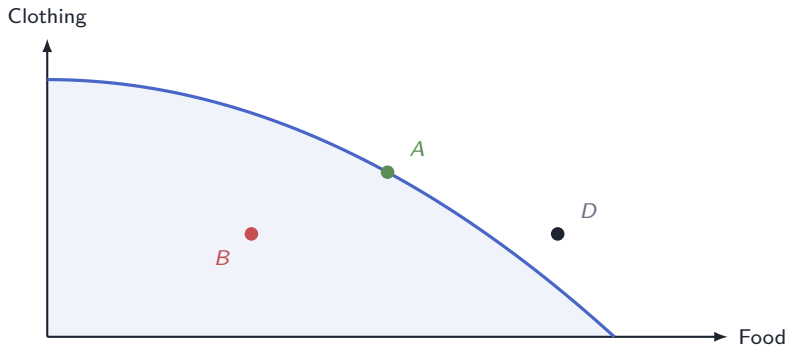
1. Find C at $F = 0, 10, 20, 30, 40$.
2. What is the maximum food it can produce?
3. Compute the cost, in clothing, of each successive 10 units of food. Is opportunity cost constant or increasing?

On paper. Four minutes.

PART 4

Efficiency, Unemployment, Growth

Three kinds of point



A on the frontier — efficient. **B inside** — inefficient. **D outside** — unattainable.

Points inside the frontier

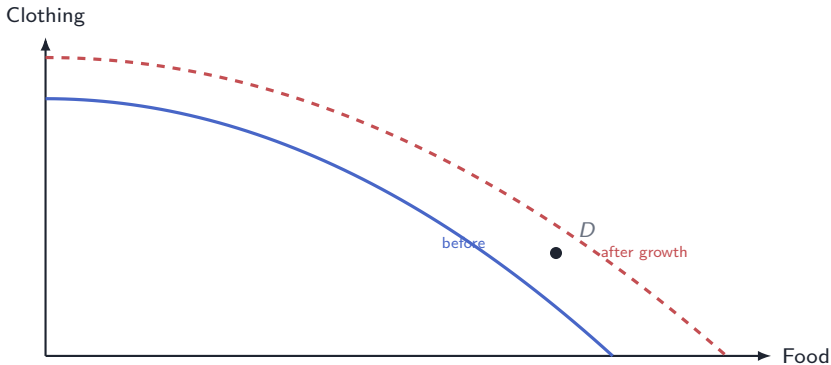
At B the economy could have **more food and more clothing** at the same time. Nothing has to be given up.

What puts an economy inside its frontier?

- **Unemployment** — workers willing to work but not working.
- Idle factories or land.
- Resources in the wrong use.

In this static model, some moves toward the frontier increase output without reducing the other good.

Economic growth shifts the frontier out



More resources, or better technology, shifts the whole frontier outward. ***D* was unattainable; now it is not.**

For each event, does the economy move **along** its frontier, move **toward** it, or does the **frontier itself shift**?

1. A recession puts 8% of workers out of work.
2. The government moves resources from clothing into food.
3. A new fertiliser doubles food output per acre.
4. Immigration raises the workforce by 10%.

Two minutes. The *why* is the marks.

PART 5

Absolute and Comparative Advantage

Two measures of the cost of a good

We now have two different ways to say a country is “better” at something.

Absolute advantage. Producing a good using *fewer resources* — here, fewer labor hours — than the other country.

From the table: Portugal needs fewer hours for *both* goods. $90 < 100$ for cloth, $80 < 120$ for wine.

Portugal has the absolute advantage in everything.

So why would Portugal trade at all?

If Portugal is better at everything, England appears to have nothing to offer.

This is where your instinct went wrong in Active Learning 1.

The answer needs a second, deeper measure — one we already have.

We met it as the slope of the frontier.

Opportunity cost, for both countries

Hours spent on one good are hours taken from the other:

$$\text{opportunity cost of a good} = \frac{\text{hours that good needs}}{\text{hours the other good needs}}$$

	1 wine costs (cloth)	1 cloth costs (wine)
England	$120 \div 100 = 1.20$	$100 \div 120 = 0.83$
Portugal	$80 \div 90 = 0.89$	$90 \div 80 = 1.125$

Within each country the two columns are reciprocals. **If one good is dear, the other must be cheap.**

Comparative advantage

Comparative advantage. Producing a good at a *lower opportunity cost* than the other country.

Wine. Portugal gives up 0.89 cloth; England gives up 1.20. $0.89 < 1.20 \Rightarrow$ **Portugal.**

Cloth. England gives up 0.83 wine; Portugal gives up 1.125. $0.83 < 1.125 \Rightarrow$ **England.**

Portugal is better at both — but it is **most better at wine**.

The key insight

Unless two countries have identical opportunity costs, each has a comparative advantage in something.

It is **impossible** for one country to have a comparative advantage in both goods — because the two opportunity costs are reciprocals. If yours is lower in one, it must be higher in the other.

That is precisely why there is always a basis for trade.

Active Learning 5

Two countries, hours of labor per unit:

	Steel	Grain
North	4	2
South	10	8

1. Who has the absolute advantage in each good?
2. Compute all four opportunity costs, showing the division.
3. Who has the comparative advantage in each good?

On paper. Five minutes.

PART 6

Specialization and the Gains

Autarky. No trade: a country consumes exactly what it produces.

Without trade, each frontier is also the *consumption* frontier.

To pin down a starting point, assume each country splits its labor equally. England 1,800 hours on each good; Portugal 1,440 on each.

A neutral choice — any point on the frontier would do.

Autarky: production equals consumption

	Cloth	Wine
England	$1800/100 = 18$	$1800/120 = 15$
Portugal	$1440/90 = 16$	$1440/80 = 18$
World	34	33

Without trade the world produces **34 cloth and 33 wine**. Remember those two numbers.

Now specialize

Each country moves its labor into its comparative-advantage good: England into cloth, Portugal into wine.

		Cloth	Wine
England	(3600/100)	36	0
Portugal	(2880/80)	0	36
World		36	36

Cloth 34 $\rightarrow$ 36. Wine 33 $\rightarrow$ 36.

Where did that come from?

The same labor, arranged differently, produces **more of both goods**.

No new resources appeared. No technology improved.

Nothing was invented and nobody worked harder. The gain came purely from **who does what**.

Full specialization is not required — any move toward the comparative-advantage good raises world output. We go all the way only to keep the arithmetic clean.

But now there is a problem

England holds 36 cloth and no wine. Portugal holds 36 wine and no cloth.

Neither country wants to consume only one good.

They need to trade. And that raises a new question: at what price?

Active Learning 6

Back to North and South. Hours per unit: North steel 4, grain 2. South steel 10, grain 8.

8. **North has 400 hours, South has 800.**

1. Compute each country's maximum steel and maximum grain.
2. With an equal labor split, what does each produce? World totals?
3. If each fully specializes in its comparative-advantage good, what are the new world totals?

On paper. Five minutes.

PART 7

The Terms of Trade

At what price?

Terms of trade. The price at which the two goods are exchanged once trade opens.

Let p be the price of one wine, measured in cloth.

Portugal sells wine. It will only accept a price *above* its own cost of making wine, 0.89.

England buys wine. It will only pay a price *below* its own cost of making wine, 1.20.

The band

$$0.89 < p < 1.20 \quad (\text{cloth per wine})$$

A deal exists only in the gap between the two opportunity costs.

Any price in that band benefits both. We take a round number in the middle, $p = 1$ — one cloth for one wine.

The price decides how the gains are split. It does not decide whether they exist.

How much to swap?

Write X for the units exchanged. England keeps $36 - X$ cloth and receives X wine; Portugal keeps $36 - X$ wine and receives X cloth.

For each to beat autarky:

$$\text{England: } 15 < X < 18, \quad \text{Portugal: } 16 < X < 18 \implies 16 < X < 18$$

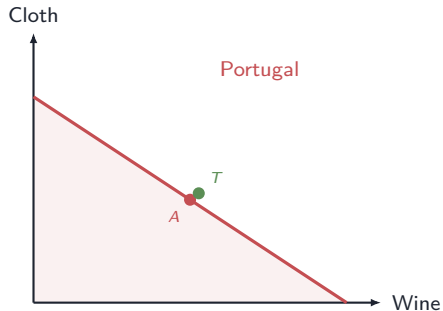
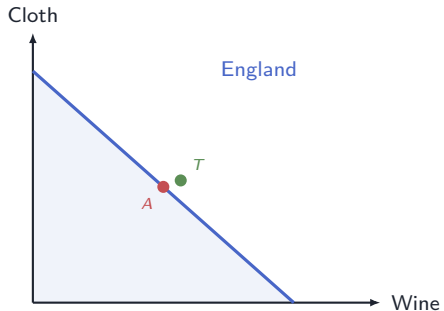
Take the midpoint, $X = 17$: England ships **17 cloth** for **17 wine**.

After trade

	Cloth made	Cloth eaten	Wine made	Wine drunk
England	36	$36 - 17 = 19$	0	$0 + 17 = 17$
Portugal	0	$0 + 17 = 17$	36	$36 - 17 = 19$
World	36	36	36	36

Each country now consumes a balanced basket it could not have produced alone.

Consumption beyond the frontier



A = autarky, on the frontier. T = after trade, beyond the frontier.

The gains, side by side

	Autarky		Trade	
	Cloth	Wine	Cloth	Wine
England	18	15	19	17
Portugal	16	18	17	19

Both countries consume more cloth *and* more wine.

Answer C was right. Trade is not a contest with a winner and a loser.

The engine of the whole result is one fact: **the two countries have different opportunity costs** — 0.89 and 1.20.

England's own wine costs 1.20 cloth. Through trade it buys wine for 1. **Cheaper.**

Portugal's own cloth costs 1.125 wine. Through trade it buys cloth for 1. **Cheaper.**

Trade is a better technology for turning what you have into what you want.

North and South again. Opportunity costs of **1 steel**: North 2 grain, South 1.25 grain.

1. Which country exports steel?
2. State the range for the terms of trade (price of 1 steel, in grain).
3. Would a price of 1 grain per steel work? What about 3?

On paper. Four minutes.

PART 8

Limits and Extensions

Four honest qualifications

Constant costs. Straight frontiers assume constant opportunity cost. With bowed frontiers — Part 3 — specialization is usually *partial*.

One factor. Labor is the only input here. A different model, Heckscher–Ohlin, adds capital and land: countries differ in what they are endowed with, and export goods that use their *abundant* factor intensively.

No trade costs. Shipping, tariffs and barriers are all set to zero.

No economies of scale. Every unit costs the same to make as the last.

Winners and losers inside a country

The model shows a gain for the *nation as a whole*. It is silent on who *inside* the country gains.

When a country opens to trade, export industries expand and import-competing industries contract. Workers in shrinking industries face unemployment and retraining costs.

Trade can raise a country's total welfare while leaving specific groups worse off.

Frictions. Real trade involves transport costs, tariffs and other barriers. These *reduce* the gains — and if they exceed the underlying cost advantage, they remove the trade altogether.

Comparative advantage is not fixed forever. It depends on relative productivity, which changes with investment, education, technology and institutions.

A country with a comparative advantage in farming today may have one in manufacturing or services tomorrow.

The principle generalized

Nothing in the argument depends on the traders being *countries*.

A lawyer types faster than her secretary — an absolute advantage in both legal work and typing.

She types 10 pages an hour and forgoes \$400 of billing, so a page costs her \$40. The secretary types 8 pages an hour and forgoes \$20 of filing, so a page costs \$2.50.

The secretary has the comparative advantage in typing — even though the lawyer types faster.

Being worse at everything is no bar to trading.

What matters is not who is better, but who gives up less. Wherever opportunity costs differ, specialization and exchange reach outcomes no country could achieve alone.

M. J. Conyon · Bentley University